

Flowcharts & Data Flow Diagrams

SmartSociety — Web-Based Housing Society Management System

Companion to Design Specification

Date August 2026

1 Resident Journey

From login to the five actions a resident can take on a given visit.

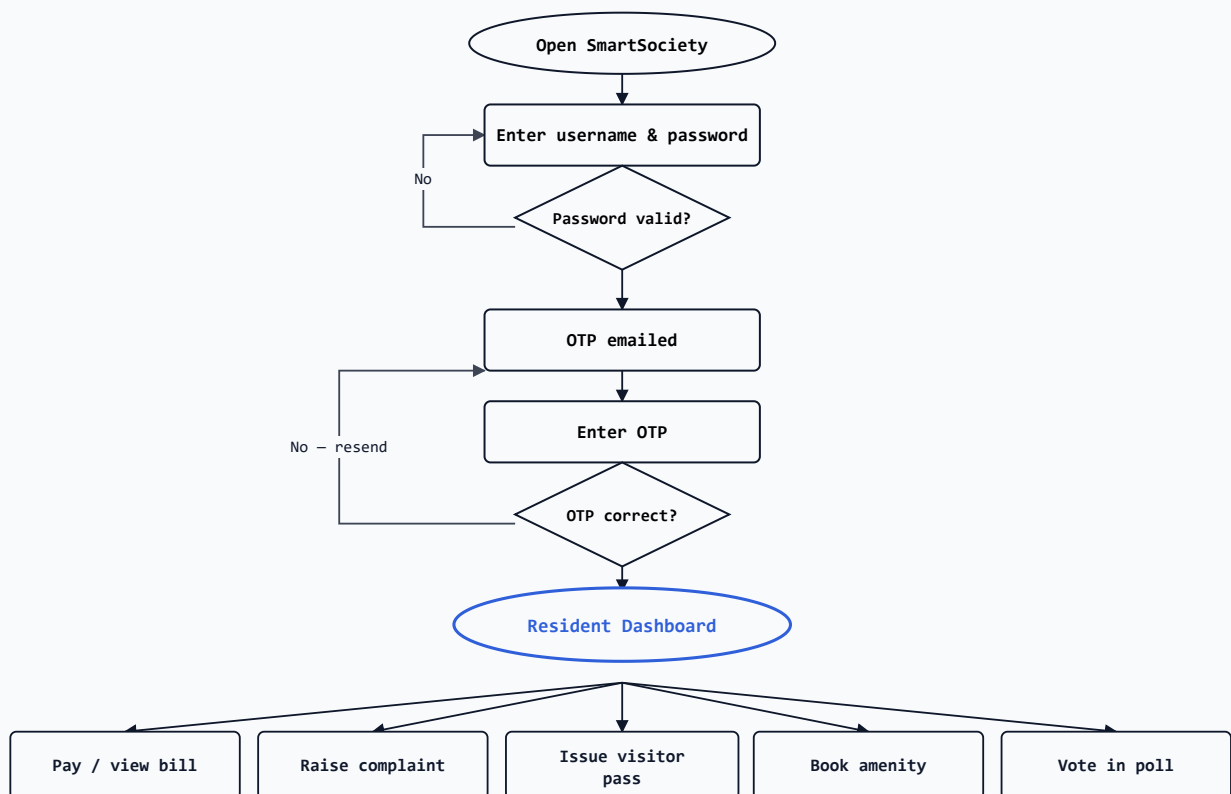


Fig. 1 – two-step OTP login, then five self-service actions the resident can take in any order. Paying a bill also offers a PDF receipt download; issuing a pass generates a QR/code shared with the guest.

2 Admin Journey

The management committee's oversight and configuration path.

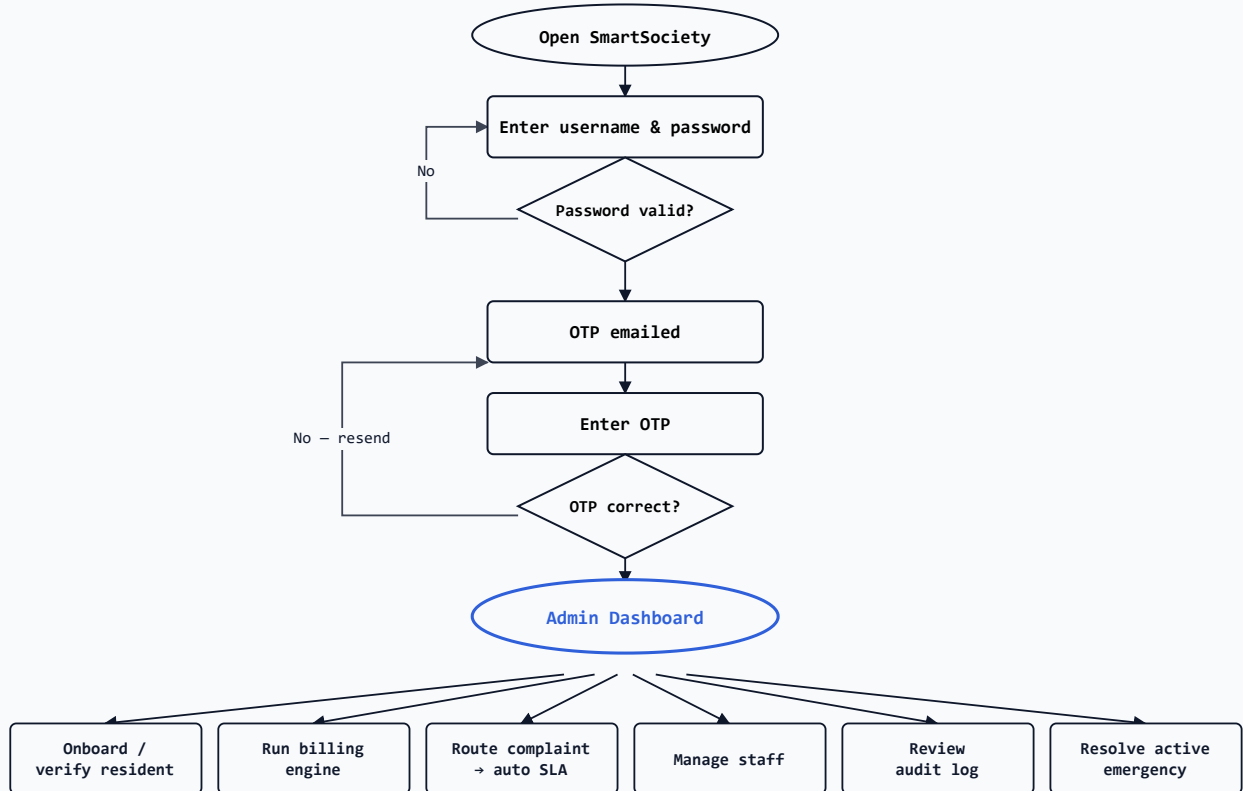


Fig. 2 – six independent oversight actions from one dashboard. Audit log review reads from the same trigger-protected table every financial edit and status change writes to (see Database Design §3).

3 Guard Journey

The security desk's gate-control loop — the part of the system where a wrong call has the most immediate consequence.

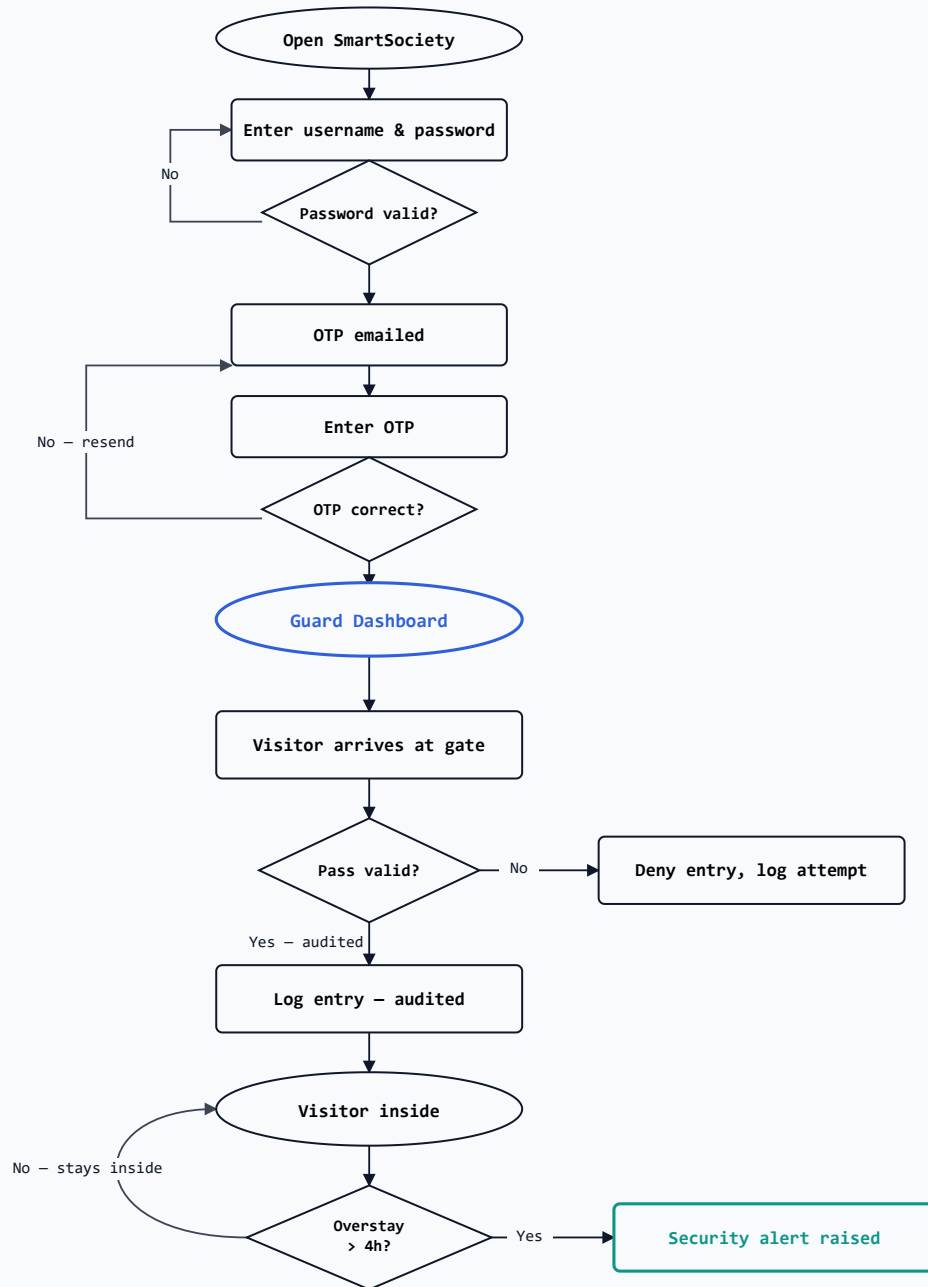


Fig. 3 – the overstay check runs as a repeating background job, not something the guard has to remember. Exit logging and the emergency-siren trigger are also available from the dashboard at any point; omitted here to keep the gate-decision loop legible.

4 Data Flow Diagram — Level 0 (Context)

The system as a single process, and the external entities it exchanges data with.

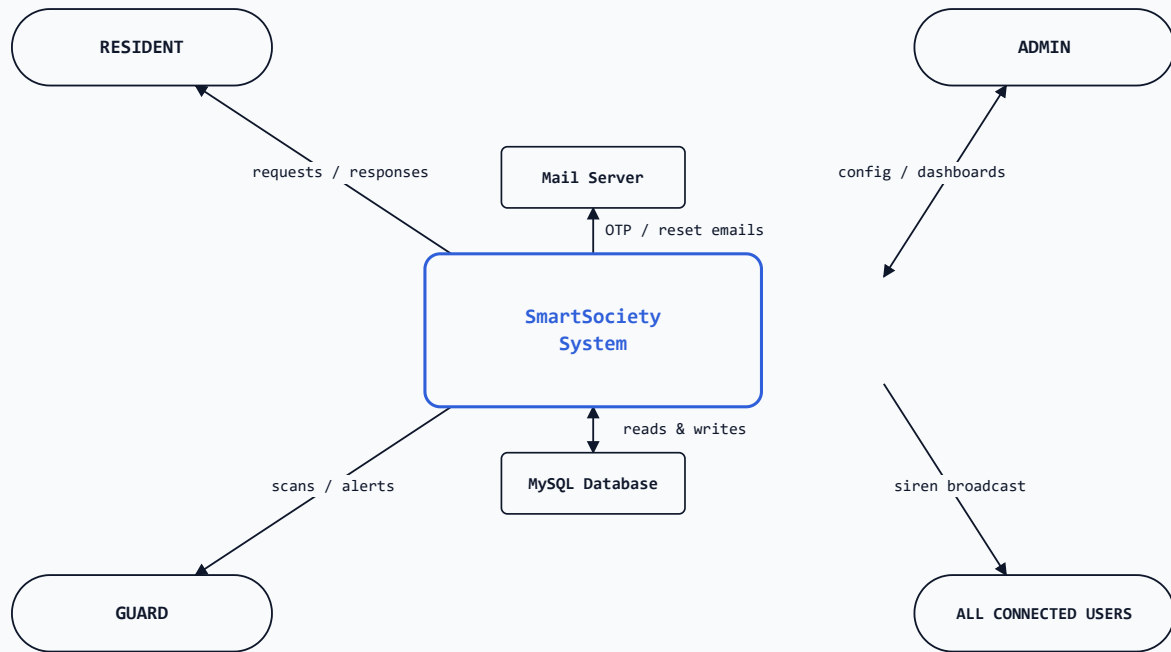


Fig. 4 – the whole application collapses to one process bounded by five external entities.

5 Data Flow Diagram — Level 1

The context process decomposed into its five core sub-processes and the stores they read/write. Laid out in three layers — external roles, processes, data stores — so every edge stays short.

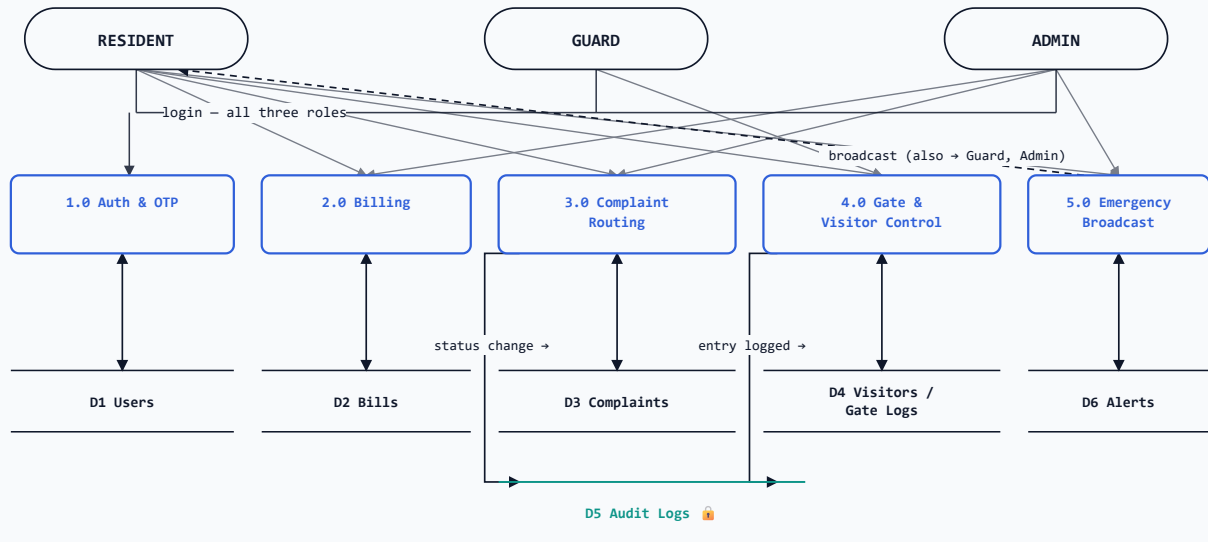


Fig. 5 – Complaint Routing and Gate & Visitor Control both write into Audit Logs (D5); D5 has no outgoing edge back into either process, reflecting the DB-trigger enforcement that makes those rows unmodifiable and undeletable once written (see Database Design §3).

6 Diagram Index

FIG.	DIAGRAM	TYPE
1	Resident Journey	Flowchart
2	Admin Journey	Flowchart
3	Guard Journey	Flowchart
4	System Context	DFD Level 0
5	Core Sub-processes	DFD Level 1

